

CSE 4345: Big Data Analytics

Week 10, Part 1 — Big Data in Real Applications: Social Media, Time Series & Health Data

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Course & Lecture Information

Lecture Identity

Course: CSE 4345
Week / Part: 10 of 11 | Part 1 of 2
CLO: CLO4
PLO: PLO(f)

CLO4

"Utilise the knowledge gained to solve big data problems and build a project applying big data tools and frameworks to real-world datasets"

Course Progression

Weeks 1–9: Foundations, storage, compute, query, pipelines, visualisation

Week 10: Applying the full stack to real application domains

Part 1 Covers

- Three application domains: social media, time series, healthcare
- **Domain A:** Social media data characteristics; community detection (Louvain); influence maximisation; sentiment analysis (VADER); trend detection
- **Domain B:** Time series properties; ARIMA; Facebook Prophet (trend + seasonality + holidays); LSTM; anomaly detection (STL, Isolation Forest); vectorised batch forecasting in Spark
- **Domain C:** Healthcare data sources (EHR, genomics, wearables, DICOM); clinical prediction models; privacy (HIPAA, PDPA Bangladesh); AlphaFold
- **Case study:** CDC syndromic surveillance — Twitter + pharmacy data → flu early warning

Lecture Agenda

- 1 Domain Overview: Theory Meets Practice
- 2 Domain A: Social Media Analytics
- 3 Domain B: Time Series Analysis
- 4 Domain C: Healthcare Big Data
- 5 Case Study: CDC Syndromic Surveillance
- 6 Student Assessment Pool

Domain Overview: Theory Meets Practice

Three Application Domains of Big Data

Social Media Analytics

- Graph-structured
- NLP / sentiment
- Community detect.
- Influence spread
- Trend detection

*Tools: GraphX, NetworkX
Neo4j, Twitter API v2*

Time Series Analysis

- Ordered, temporal
- Seasonal, noisy
- ARIMA
- Prophet (Facebook)
- LSTM, anomaly det.

*Tools: Prophet, Spark
MLlib, statsmodels*

Healthcare Big Data

- EHR records
- Genomics (omics)
- Wearables (IoT)
- Medical imaging
- Privacy critical

*Tools: FHIR, TensorFlow
HAPI, Spark MLlib*

Shared Stack: HDFS / S3 | Kafka | Spark | Hive | Grafana / Superset

Domain A: Social Media Analytics

Four Data Dimensions of Social Media

- ➊ **Graph-structured:** Platforms are networks. Twitter/X has 41.7 million users (Kwak et al., 2010); Facebook has 3 billion. The graph (who follows whom, who retweets whom) is the primary analytical object — not the individual post.
- ➋ **Temporal:** Every post, like, and share carries a timestamp. Events cascade through the network over minutes to hours. Time is a first-class dimension.
- ➌ **Text (NLP):** Post content is unstructured natural language — short, noisy, abbreviation-heavy, and multilingual. Standard NLP pipelines must handle slang, emojis, and code-switching (Bangla–English mixing).
- ➍ **Multimedia:** Images and videos dominate modern social media (Instagram, TikTok). Computer vision (object detection, OCR) is needed to analyse this content.

Social Media Data: Characteristics and Scale

Scale Numbers (2024)

Platform	Posts/day
Twitter / X	500 M tweets
Facebook	100 B interactions
Instagram	100 M photos
TikTok	1 B videos viewed
YouTube	500 hours uploaded/min

Bangladesh context:

Facebook is the dominant social platform (50+ million users). During national disasters (floods, cyclones), crisis information spreads via Facebook Pages before official government channels. Misinformation analysis on Bangla-language Facebook posts is an active NLP research area.

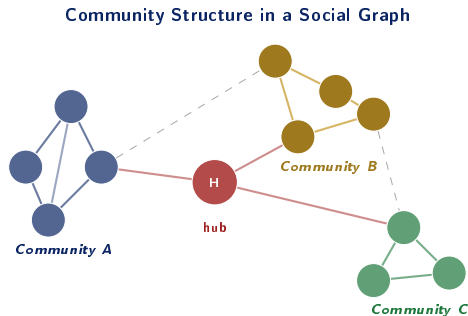
Community Detection: Louvain Algorithm

A **community** in a social graph is a group of nodes more densely connected to each other than to the rest of the graph. Detecting communities reveals interest groups, political factions, professional clusters, or coordinated bot networks. **Louvain algorithm** (Blondel et al., 2008):

- 1 Start: each node is its own community
- 2 Greedily merge each node into the neighbour community that maximises **modularity** (fraction of edges within communities minus expected fraction by chance)
- 3 Aggregate merged communities into super-nodes; repeat
- 4 Terminate when modularity gain is zero

Complexity: $O(n \log n)$ — scalable to billion-node graphs in Spark GraphX.

Community Detection and Influence Maximisation



Influence Maximisation

Which k nodes to seed so that information (or a product promotion) spreads to the maximum number of users? Kempe et al. (KDD 2003 Best Paper) proved this is NP-hard but gave a greedy approximation with a $(1 - 1/e)$ guarantee.

Sentiment Analysis: VADER

VADER (Valence Aware Dictionary for Sentiment Reasoning, Hutto & Gilbert 2014) is a lexicon-and-rule-based sentiment analyser tuned specifically for **social media text**. It handles:

- Capitalisation: GREAT > great
- Punctuation: great!!! > great
- Degree modifiers: very good, kind of bad
- Negation: not good → negative
- Emojis: :) → positive score

Output: compound score $\in [-1, +1]$; positive (> 0.05), neutral, negative (< -0.05). **Spark pipeline:**

Apply VADER to each tweet via a UDF on a DataFrame. Aggregate sentiment scores by time window and topic. Push to Grafana for the streaming dashboard built in Week 9.

Sentiment Analysis and Trend Detection

Trend Detection

Twitter's trending topics algorithm uses **Locality-Sensitive Hashing (LSH)** on character n -grams to detect sudden spikes in the frequency of a phrase *relative to its historical baseline*. A phrase that normally appears 10 times/hour and suddenly appears 10,000 times/hour is trending — regardless of absolute volume.

Bangladesh application: During Eid-ul-Adha 2024, the hashtag #QurbaniEid showed a 1,200× spike on X (Twitter) over 6 hours. A sentiment breakdown using VADER and a Bangla lexicon revealed 92% positive sentiment, confirming organic celebration rather than coordinated activity.

Ethical Risk

Sentiment analysis on political speech can be used to **suppress dissent** (flagging negative sentiment about a government policy) or **microtarget advertising** at vulnerable users. Transparency and purpose limitation are essential.

Domain B: Time Series Analysis

Time Series: Properties and Classical Forecasting (ARIMA)

What Makes Time Series Special

Time series data is fundamentally different from i.i.d. tabular data:

- **Order matters:** shuffling rows destroys the signal. Timestamps must be preserved.
- **Autocorrelation:** today's value is correlated with yesterday's. Standard regression that assumes independent observations is invalid.
- **Seasonality:** regular cyclical patterns (daily, weekly, annual) must be modelled explicitly.
- **Trend:** long-run upward or downward drift independent of seasonality.
- **Non-stationarity:** mean and variance may change over time. Most models require stationary input (achieved via differencing).

ARIMA: Classical Statistical Forecasting

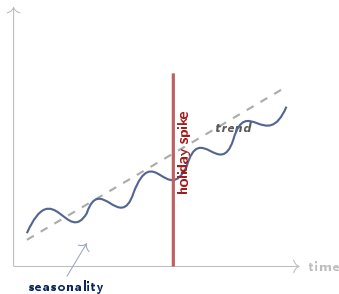
ARIMA(p, d, q) = AutoRegressive Integrated Moving Average:

- **AR**(p): current value depends on past p values
- **I**(d): difference the series d times to achieve stationarity
- **MA**(q): current value depends on past q forecast errors

Best for: short stationary series, no strong seasonality, small datasets.

Time Series: Properties and Classical Forecasting (ARIMA)

Typical Business Time Series



ARIMA Limitations

ARIMA assumes **stationarity**, struggles with complex multi-seasonal patterns, and requires expert parameter selection (p , d , q). At big data scale (10,000 time series), it cannot be manually fitted.

Facebook Prophet: Decomposable Forecasting at Scale

Prophet's Additive Model (Taylor & Letham, 2018)

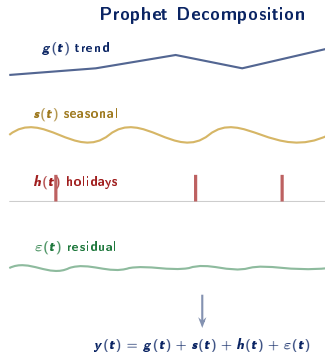
Prophet decomposes a time series into interpretable components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t)$$

- $g(t)$: **trend** — piecewise linear or logistic growth; automatically detects **change points** where the trend shifts
- $s(t)$: **seasonality** — Fourier series capturing annual, weekly, and daily cycles; handles multiple simultaneous seasonalities
- $h(t)$: **holidays and events** — user-specified list of irregular one-off events (Eid, Puja, election days, cricket World Cup) that cause known anomalies
- $\varepsilon(t)$: residual noise (assumed i.i.d.)

Why it scales: Prophet fits each series *independently* using Stan (probabilistic programming). Running 10,000 series is embarrassingly parallel — one Spark partition per series via `applyInPandas`.

Facebook Prophet: Decomposable Forecasting at Scale



Anomaly Detection in Time Series

An **anomaly** (outlier, change point, or contextual deviation) is a data point that does not conform to the expected pattern. Two methods:

- 1 **STL Decomposition** (Seasonal-Trend decomposition using LOESS): Decompose the series into trend, seasonal, and residual. Flag residuals whose absolute value exceeds $k \times \text{IQR}$ of residuals as anomalies. Interpretable; works well for regular seasonality.
- 2 **Isolation Forest**: Ensemble of random trees that *isolates* anomalies by partitioning the feature space. An anomaly requires fewer splits to isolate $\Rightarrow$ shorter path length. No assumption about data distribution; scales to millions of observations in Spark MLlib.

Bangladesh application: bKash monitors transaction volume per mobile agent per hour. An agent whose hourly volume jumps from 10 transactions to 10,000 is flagged by Isolation Forest as a potential money-laundering node.

Vectorised Batch Forecasting in Spark

Problem: A retail chain needs 90-day demand forecasts for 5,000 store-product SKU combinations. Fitting 5,000 separate Prophet models sequentially on one machine takes 8+ hours. **Spark solution:**

- 1 Load all SKU time series into a Spark DataFrame (one row per day per SKU)
- 2 Group by `sku_id`: each group is one independent series
- 3 Apply `applyInPandas(prophet_fit_and_forecast)`: Spark sends each SKU's data to one partition; Prophet runs inside a Pandas UDF
- 4 All 5,000 models fit in parallel across the cluster
- 5 Results written to Hive/S3 as Parquet; served to dashboard

Runtime: 8 hours $\rightarrow$ $\sim$ 12 minutes on a 20-executor Spark cluster ($40\times$ speedup).

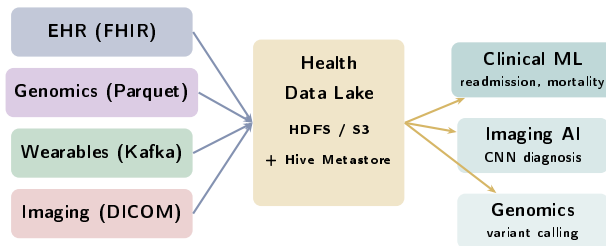
Domain C: Healthcare Big Data

Healthcare Data Sources and Characteristics

Four Healthcare Data Modalities

- ➊ **Electronic Health Records (EHR):** structured diagnoses (ICD-10), medications, lab results, vital signs, clinical notes (unstructured free text). Semi-structured: HL7 FHIR resources (Week 2).
- ➋ **Genomics (Omics):** A single whole-genome sequence is ~ 200 GB. A clinical genomics study of 10,000 patients = 2 PB. Requires specialised distributed storage (Apache Parquet + GATK pipeline on Spark).
- ➌ **Wearables and IoT:** Continuous heart rate, SpO_2 , sleep, glucose monitor readings. Generates time-series data at per-second resolution; requires streaming ingestion (Kafka) and anomaly detection.
- ➍ **Medical Imaging (DICOM):** CT, MRI, X-ray, pathology slides are unstructured binary files (200 MB – 2 GB each). Stored in PACS; analysed by deep learning (CNN, ViT) not SQL.

Healthcare Data Sources and Characteristics



EHR-Based Clinical Prediction (Obermeyer & Emanuel, 2016)

The landmark NEJM paper argued that **ML models trained on EHR big data** systematically outperform physician heuristics for three critical clinical tasks:

- **ICU mortality prediction:** Random forest trained on vitals, labs, and prior diagnoses achieves AUROC > 0.90 — identifying patients who will die within 24 hours earlier than conventional scoring (APACHE, SOFA)
- **30-day hospital readmission:** Gradient boosting on discharge records identifies high-risk patients for post-discharge follow-up, reducing readmission rates by 20%
- **Sepsis early warning:** Continuous streaming of EHR vitals through a real-time model (Spark Streaming) detects sepsis onset 6 hours before clinical criteria are met

Data requirement: 1 million+ patient records for reliable generalisation. Bangladesh's DGHS has 15+ years of digitised records at national hospitals — sufficient scale if integrated.

Privacy: HIPAA, GDPR, and Bangladesh

Regulation	Key Requirement
HIPAA (USA)	De-identify 18 PHI identifiers before research use; consent required
GDPR (EU)	Data minimisation; purpose limitation; right to erasure
PDPA Bangladesh	Under development; based on GDPR principles; NID-linked records require explicit consent

De-identification techniques:

- **Safe harbour:** remove 18 specific identifiers (name, DOB, zip, NID, phone, etc.)
- **Expert determination:** statistical disclosure analysis by qualified expert
- **Differential privacy:** add calibrated noise to aggregate statistics so individual records cannot be inferred
- **Federated learning:** train models at each hospital without moving raw data

The 50-Year Protein Folding Problem

A protein's **3D structure** determines its function. From an amino acid sequence, predicting the 3D fold was considered one of biology's hardest problems. Experimental determination (X-ray crystallography) takes months per protein and costs thousands of dollars. **AlphaFold 2 (DeepMind, 2020):**

- Trained on the **Protein Data Bank**: 170,000 experimentally determined structures
- Architecture: **Evoformer** transformer block applied to multiple sequence alignments (MSAs) across evolutionarily related proteins
- Achieves median GDT score $>92\%$ (expert-level accuracy)
- **2022**: Released 200 million predicted protein structures in the public AlphaFold Protein Structure Database — effectively solving protein structure prediction for nearly the entire known proteome

AlphaFold represents the most dramatic impact of big data + deep learning on experimental science in the last decade.

Why This Is a Big Data Problem

The input to AlphaFold is not just the query sequence — it is a **multiple sequence alignment (MSA)** of thousands of evolutionarily related sequences from millions of organism genomes:

- UniProt database: 250 million protein sequences (>200 GB compressed)
- MSA generation for one protein query: search 250 M sequences, align top hits, encode as a 2D matrix
- Training: required 128 TPU v3 cores for several weeks on the PDB + UniRef90

Bangladesh relevance: ICDDR,B and DGHS are using AlphaFold to study *Vibrio cholerae* protein structures relevant to cholera vaccine design — leveraging a globally trained model on locally relevant pathogens.

Case Study: CDC Syndromic Surveillance

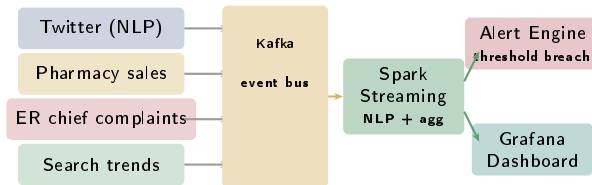
The Problem: Surveillance Lag

Traditional disease surveillance relies on **physician-reported cases** submitted to the CDC. The lag between a patient's first symptoms and the CDC receiving the report is **1–2 weeks**. By then, an outbreak may be exponentially growing. **Syndromic surveillance** uses **non-traditional data sources** that reflect illness *before* a diagnosis is made:

- **Social media:** Twitter/X posts containing words “flu”, “fever”, “can’t taste” peak 7–10 days *before* ER visits
- **Pharmacy sales:** OTC antipyretic (paracetamol, ibuprofen) purchase spikes at 3–5 days before clinical confirmation
- **ER chief complaints:** text from triage notes processed with NLP to extract “influenza-like illness” concepts in real time
- **Search query volume:** Google search trends for flu symptoms correlate with 1-week-leading incidence (Google Flu Trends – discontinued due to accuracy issues)

CDC Syndromic Surveillance: Big Data for Epidemic Early Warning

Syndromic Surveillance Pipeline



Lesson: Supplement, Not Replace

Google Flu Trends overfit to search behaviour and **over-predicted** flu incidence by $2\times$ in 2013. Syndromic surveillance provides **early signals**, not ground truth — it must be combined with traditional clinical reporting.

Student Assessment Pool

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. Facebook's Prophet library is primarily designed for:

- ☐ (a) Graph analytics and community detection in social networks
- ☐ (b) Real-time stream processing of events
- ☒ (c) **Time series forecasting with decomposable trend, seasonality, and holiday components**
- ☐ (d) Text sentiment classification using deep learning

Q2. Which algorithm is most commonly used for **community detection** in large-scale social network graphs?

- ☒ (a) K-Means clustering
- ☐ (b) PageRank
- ☐ (c) **Louvain algorithm**
- ☐ (d) ARIMA

Instructor Notes

Q1: Distractor (d) — Prophet has nothing to do with sentiment; it is a decomposable time series model. Distractor (a) — graph analytics uses GraphX/NetworkX, not Prophet. **Q2:** PageRank (b) measures node *importance*, not community membership. K-Means (a) clusters vectors, not graph topologies. ARIMA (d) is a time series method, leaving (c) as the only remaining candidate for community detection.

Q3. HIPAA in the United States primarily regulates the privacy and security of:

- ☐ a) Financial transaction data and bank records
- ☒ b) **Healthcare data including patient records and medical information**
- ☐ c) Social media user data and behavioural profiles
- ☐ d) Government data retention and public records

Q4. DeepMind's AlphaFold used big data and deep learning to solve:

- ☐ a) Real-time financial fraud detection at bank-card transaction scale
- ☐ b) Social network influence maximisation over billion-node graphs
- ☒ c) **The protein structure prediction problem, predicting 3D structure from amino acid sequence**
- ☐ d) Time series anomaly detection in IoT sensor streams

Instructor Notes

Q3: A common confusion — HIPAA (Health Insurance Portability and Accountability Act) is specifically for healthcare. Financial data is regulated by GLBA (US) or Basel III. **Q4:** AlphaFold's achievement was predicting protein 3D structure from sequence alone, using MSAs from 250 M sequences and training on the 170,000 experimentally solved structures in the Protein Data Bank. It won the 2024 Nobel Prize in Chemistry.

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	Answer
1. Google Flu Trends successfully replaced traditional CDC flu surveillance and has been used as the primary source ever since.	False
2. Time series data is order-sensitive: randomly shuffling the records in a time series changes the analysis result.	True
3. Social network graphs are typically sparse, with most nodes connected to only a small fraction of all possible neighbours.	True
4. Medical imaging data such as MRI and CT scans is classified as structured data because it is stored in files.	False

Instructions

Answer each question in **100–150 words**.

- D1.** Describe **three types of analysis** commonly performed on social media data (e.g., community detection, sentiment analysis, influence maximisation). For each, state one specific **ethical risk** that arises when the analysis is deployed at scale.
- D2.** What is the difference between **ARIMA** and **Facebook Prophet** for time series forecasting? Under what conditions would you prefer Prophet over ARIMA? When might ARIMA still be the better choice?
- D3.** Identify **three data integration and privacy challenges** specific to healthcare big data analytics. For each, describe one technical mitigation strategy (e.g., differential privacy, federated learning, de-identification).

Instructions

Answer in **200–300 words** with end-to-end pipeline reasoning.

A1. Flu Outbreak Early-Warning System for Bangladesh:

Design a big data system to detect flu outbreaks in Bangladesh using Twitter/Facebook posts and pharmacy purchase records from major pharmacy chains.

- (a) How would you **collect and ingest** these two data sources? (Specify tools: Kafka, Sqoop, Flume, REST API)
- (b) What **features** would you extract from each source for a prediction model? (Name specific NLP features and pharmacy signal metrics)
- (c) What **model** would you use, and how would you validate it against ground-truth CDC/DGHS surveillance data?
- (d) What are the **ethical risks** of this system? How would you mitigate them?

A2. Retail Demand Forecasting Pipeline:

A large Bangladeshi supermarket chain needs 90-day demand forecasts for 5,000 store–product SKU combinations. Design the **end-to-end big data pipeline**: (a) data ingestion from POS systems; (b) storage and partitioning strategy in HDFS/S3; (c) forecasting model choice (ARIMA

Textbook & Reading References

Primary Textbooks for Week 10

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 11**
 - Ch. 11: Big Data Use Cases — social media, healthcare, retail, government analytics
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Ch. 13**
 - Ch. 13: Big Data in Practice — domain-specific deployments and lessons learned
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed. **Ch. 10**
 - Ch. 10: Social-Network Graphs — community structure, influence propagation, PageRank

Research Articles (covered in Week 10, Part 2)

- Kwak, H. et al. (2010). **What is Twitter, a Social Network or a News Media?** *WWW 2010*.
- Taylor, S. J., & Letham, B. (2018). **Forecasting at Scale.** *The American Statistician*. [Facebook Prophet paper]
- Obermeyer, Z., & Emanuel, E. J. (2016). **Predicting the Future — Big Data, Machine Learning, and Clinical Medicine.** *New England Journal of Medicine*.
- Jumper, J. et al. (DeepMind, 2021). **Highly Accurate Protein Structure Prediction with AlphaFold.** *Nature*. [DOI: 10.1038/s41586-021-03071-3]

Questions & Discussion

Week 10, Part 1 | CSE 4345 Big Data Analytics

“Big data is not about the data. It is about the decisions.”

— Kenneth Cukier & Viktor Mayer-Schönberger, *Big Data: A Revolution*, 2013

Next Lecture: Week 10, Part 2

Research Articles: Kwak et al. (2010) Twitter as news medium · Taylor & Letham (2018) Prophet · Obermeyer & Emanuel (2016) clinical ML

Case Study: **CDC Syndromic Surveillance** — integrating Twitter, pharmacy, and ER data to detect flu outbreaks 1–2 weeks before clinical confirmation

Reading before Part 2: [LRU] Ch.10 · Taylor & Letham (2018) Prophet paper (open access)